

Z-residuals for Checking Bayesian Hurdle Models

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Section 1

Introduction

Residual diagnostics plays very important roles in frequentist statistics:

- Assessing the normality of residuals with graphical tools, such as scatterplot, QQ plot, etc.
- Using residual plots for discovering the problem of a fitted model, such as wrong distributional family, non-linear covariate effect, and non-constant variance.
- Identifying outliers
- Overall goodness-of-fit (GOF) and other statistical tests for quantifying the inadequacy of a fitted model

- Residual diagnostics is not a common practice in Bayesian statistics
- Recently, Z-residuals have been developed to assess the goodness-of-fit (GOF) of various models, such as survival models and count regression models in frequentist frameworks.
- In this talk, we extend the application of Z-residual to Bayesian hurdle models.

Section 2

Hurdle Models

Hurdle Models

- Hurdle models can be viewed as a two-component mixture model consisting of
 1. a zero proportion, and
 2. the positive observations component following a truncated count distribution, such as a truncated Poisson or truncated negative binomial (NB) distribution.
- The probability function of hurdle models is defined as

$$Pr(Y_i = y_i) = \begin{cases} p_i, & y_i = 0, \\ (1 - p_i) \frac{p(y_i; \mu_i)}{1 - p(y_i = 0; \mu_i)}, & y_i > 0, \end{cases} \quad (1)$$

where p_i is the proportion of zeros, $p(y_i; \mu_i)$ represents a PMF for a regular count distribution with a vector of parameters μ_i , and $p(y_i = 0; \mu_i)$ is the distribution evaluated at zero.

Examples of Hurdle Models

- We will focus on the hurdle negative binomial (HNB) and hurdle Poisson (HP) model. The probability function of HNB model is defined as:

$$Pr(Y_i = y_i) = \begin{cases} p_i, & y_i = 0, \\ \frac{1-p_i}{1-\left(\frac{k}{\mu_i+k}\right)^k} \frac{\Gamma(y_i+k)}{\Gamma(k)y_i!} \left(\frac{\mu_i}{\mu_i+k}\right)^{y_i} \left(\frac{k}{\mu_i+k}\right)^k, & y_i > 0, \end{cases} \quad (2)$$

where μ_i is the mean parameter, k is the dispersion parameter of NB distribution.

- The probability function of HP model is defined as:

$$Pr(Y_i = y_i) = \begin{cases} p_i, & y_i = 0, \\ (1-p_i) \frac{e^{-\mu_i} \mu_i^{y_i} / y_i!}{1-e^{-\mu_i}}, & y_i > 0, \end{cases} \quad (3)$$

where μ_i is the mean parameter of Poisson distribution.

Examples of Hurdle Models (Cont'd)

- The hurdle model is generated as follows:
 - Count model: $\log(\mu_i) = \mathbf{x}_i^T \boldsymbol{\alpha}$
 - Zero model: $\text{logit}(p_i) = \mathbf{z}_i^T \boldsymbol{\beta}$
 - where μ_i is the mean, p_i is the probability of zeros, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are regression coefficients for the covariates $\mathbf{x}_i$ and $\mathbf{z}_i$.

Section 3

Z-residuals

Z-residuals given True Parameters

- The **middle-value predictive p-value (MPP)** is defined in the literature (Marshall and Spiegelhalter 2007 and others):

$$\text{mpp}(y_i^{\text{obs}}|\theta) = Pr(y_i > y_i^{\text{obs}}|\theta) + 0.5Pr(y_i = y_i^{\text{obs}}|\theta). \quad (4)$$

- The **randomized predictive p-value (RPP)** is defined in Feng, Li, and Sadeghpour (2020):

$$\text{rpp}(y_i^{\text{obs}}|\theta) = Pr(y_i > y_i^{\text{obs}}|\theta) + U_i Pr(y_i = y_i^{\text{obs}}|\theta), \quad (5)$$

where $U_i \sim \text{Uniform}(0, 1)$.

- Under the true model, the RPP is uniformly distributed over $(0, 1)$.

Z-residuals given True Parameters (cont'd)

- To compute the **Z-residuals**, we can transform $\text{rpp}(y_i^{\text{obs}}|\theta)$ with the negative normal quantile:

$$z_i(y_i^{\text{obs}}|\theta) = -\Phi^{-1}(\text{rpp}(y_i^{\text{obs}}|\theta)), \quad (6)$$

where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of a standard normal distribution.

- Under the true model, the Z-residual is distributed as standard normal.

- The **posterior RPP** $\text{rpp}^{\text{Post}}(y_i^{\text{obs}})$ is defined based on the full data posterior density of parameters θ given observations $\mathbf{y}_{1:n}^{\text{obs}}$.
- The posterior RPP is computed given the Markov Chain Monte Carlo (MCMC) samples, and denoted by $\widehat{\text{rpp}}^{\text{Post}}(y_i^{\text{obs}})$.
- The **posterior RPP Z-residual** can be defined as

$$z_i^{\text{Post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{Post}}(y_i^{\text{obs}})). \quad (7)$$

- The posterior MPP and posterior MPP Z-residuals can be defined in the same way.
- We will occasionally use ‘Post.’ to abbreviate the term ‘posterior’.

Importance-sampling CV Z-residuals

- The **Importance-sampling Cross-validatory (ISCV) RPP** $\text{rpp}^{\text{ISCV}}(y_i^{\text{obs}} | \mathbf{y}_{-i}^{\text{obs}})$ is defined based on the cross-validatory (CV) distribution, and the Importance Sampling method.
- The ISCV RPP is computed given the MCMC samples $\{(\theta^{(t)}); t = 1, \dots, T\}$, and denoted by:

$$\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}}) = \frac{\sum_{t=1}^T [\text{rpp}(y_i^{\text{obs}} | \theta^{(t)}) w_i^{\text{IS}}(\theta^{(t)})] / T}{\sum_t w_i^{\text{IS}}(\theta^{(t)}) / T}, \quad (8)$$

where $w_i^{\text{IS}}(\theta)$ is the ratio:

$$w_i^{\text{IS}}(\theta) = \frac{P_{\text{post}(-i)}(\theta | y_{-i}^{\text{obs}})}{P_{\text{post}}(\theta | y_{1:n}^{\text{obs}})} \times \frac{C_2}{C_1} = \frac{1}{P_y(y_i^{\text{obs}} | \theta)} \quad (9)$$

- The **ISCV RPP Z-residual** can be defined as

$$z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}})). \quad (10)$$

- Same to way define the ISCV MPP Z-residuals.

Posterior RPP and ISCV RPP: Histogram

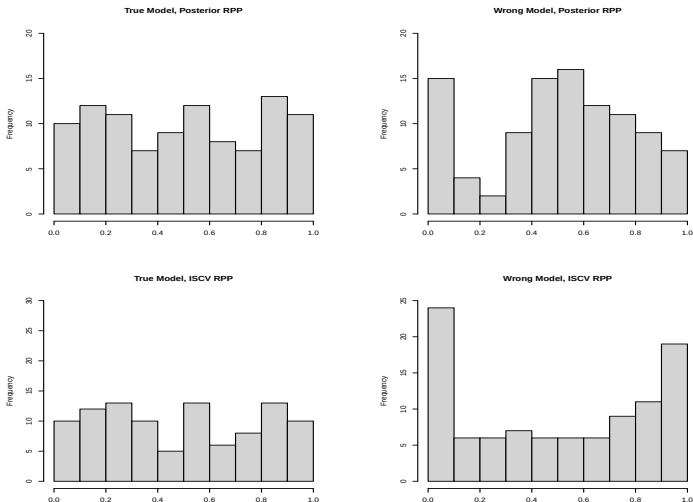


Figure 1: Assessment of uniform distribution for posterior RPP and ISCV RPP under the true model: HNB (left) and wrong model: HP (right). Sample size of $n = 100$ and zero proportions of $p = 30\%$.

Summary

We have developed generic R functions to calculate these four types of predictive p -values for different types of models, specifically designed for outputs from the `brms` package.

Z-residual Name	Formula
Post. RPP	$z_i^{\text{post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{post}}(y_i^{\text{obs}}))$
Post. MPP	$z_i^{\text{post}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{mpp}}^{\text{post}}(y_i^{\text{obs}}))$
ISCV RPP	$z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{rpp}}^{\text{ISCV}}(y_i^{\text{obs}}))$
ISCV MPP	$z_i^{\text{ISCV}}(y_i^{\text{obs}}) = -\Phi^{-1}(\widehat{\text{mpp}}^{\text{ISCV}}(y_i^{\text{obs}}))$

Table 1: Four types of Z-residuals

Section 4

Simulation Study I: Detection of Mis-specified Distribution Family

Subsection 1

Data Generation and Model Fitting

Data Generation and Model Fitting

The data is generated as follows:

- A covariate: $X_i \sim N(0, 1)$.
- Five noises: $Z_i^{(j)} \sim N(0, 1), j = 1, \dots, 5$.
- Regression coefficients: $\alpha_0 = 2, \alpha_1 = 8, \beta_0 = -1, \beta_1 = -1$.
- Count model: $\log(\mu_i) = \alpha_0 + \alpha_1 X_i$.
- Zero model: $\text{logit}(p_i) = \beta_0 + \beta_1 X_i$.
- The zero indicator: $O_i \sim \text{Binom}(1, p_i)$.
- The count response: $y_i = (1 - O_i) \times C_i$,
where $C_i \sim \text{Truncated NB}(\mu_i, \gamma, \pi), \gamma = 4$ is the dispersion parameter,
and $\pi = \gamma/(\gamma + \mu_i)$ is the probability.
- The above steps are repeated for all simulated datasets. In total, we generate 600 response variables from the HNB model with sample sizes of $n = 50, 100, 200$ and zero proportions of $p = 30\%$.

Data Generation and Model Fitting (cont'd)

We fit four candidate models to the simulated datasets utilizing the function `brm()` from R package `brms`. The models are fitted as listed below:

- **HNB Model (true model):**

$$y_i \sim \text{HNB}(\mu_i, \gamma, p_i),$$
$$\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j},$$
$$\text{logit}(p_i) = \beta_0 + \beta_1 X_i + \sum_{j=1}^5 b_j Z_{i,j};$$

- **NB Model (wrong model):**

$$y_i \sim \text{NB}(\mu_i, \gamma),$$
$$\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j};$$

- **HP Model (wrong model):**

$$y_i \sim \text{HP}(\mu_i, p_i),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j},$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 X_i + \sum_{j=1}^5 b_j Z_{i,j};$$

- **Poisson Model (wrong model):**

$$y_i \sim \text{Poisson}(\mu_i),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_i + \sum_{j=1}^5 a_j Z_{i,j}.$$

- We calculate four different kinds of Z-residuals and compare them.
- This simulation study focuses on comparing the performance between RPP Z-residuals and MPP Z-residuals.
- We provide both graphical (scatter plots, Q-Q plots) and numerical (Shapiro-Wilk test for checking normality) results for Z-residuals.

Subsection 2

Z-residual Diagnostics

Posterior RPP vs. Posterior MPP Z-residual: Scatter Plot

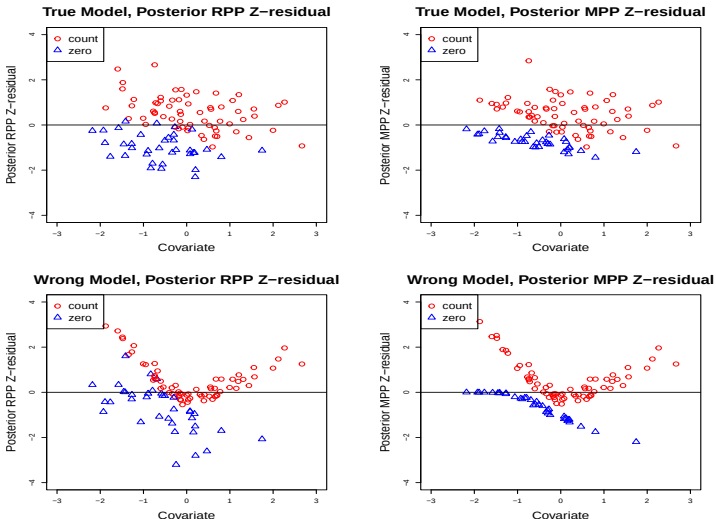


Figure 2: Posterior RPP Z-residuals and posterior MPP Z-residuals vs. covariate for HNB and NB models when the dataset is simulated from HNB ($n = 100, p = 30\%$).

Posterior RPP vs. Posterior MPP Z-residual: Q-Q Plot

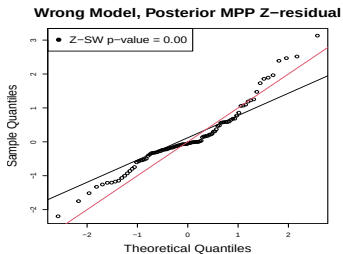
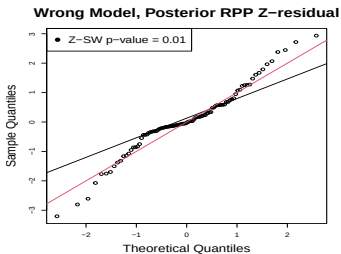
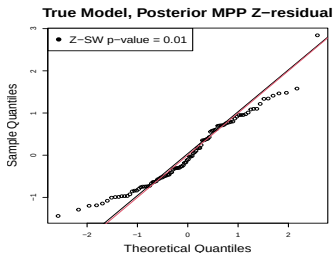
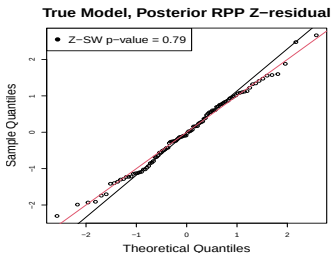


Figure 3: Q-Q plots for posterior RPP Z-residuals vs. posterior MPP Z-residuals for HNB and NB models when the dataset is simulated from HNB ($n = 100, p = 30\%$).

Subsection 3

Type-I Error Rates and Powers of SW Tests

Model Rejection Rates

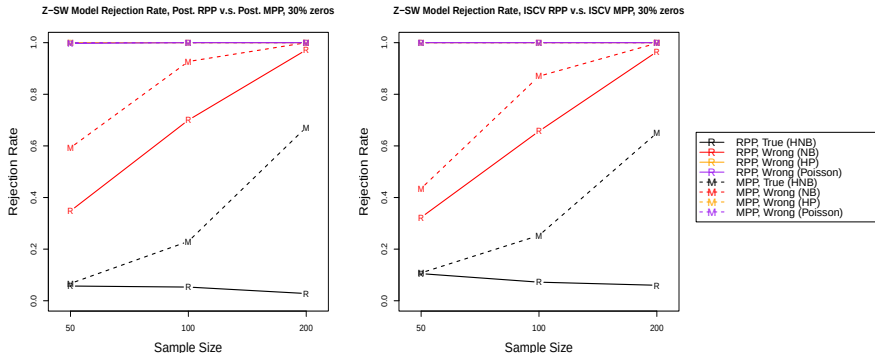


Figure 4: Model rejection rates based on Shapiro-Wilk (SW) tests of RPP vs. MPP Z-residuals when the dataset is simulated from HNB. A model is rejected when the test p -value is smaller than 5%. The number of replications is 600. The zero proportion is $p = 30\%$.

Section 5

Simulation Study II: Detection of Functional Form of Covariates

Subsection 1

Data Generation and Model Fitting

Data Generation and Model Fitting

We add the logarithmic covariate effect to $X_{i,1}$ in the count model.

- Two covariates: $X_{i,1} \sim N(0, 15)$, $X_{i,2} \sim \text{Binom}(1, 0.5)$.
- Five noises: $Z_i^{(j)} \sim N(0, 1), j = 1, \dots, 5$.
- Regression coefficients:
 $\alpha_0 = -1, \alpha_1 = 6, \alpha_2 = 1, \beta_0 = -1, \beta_1 = -1, \beta_2 = -5$.
- Count model: $\log(\mu_i) = \alpha_0 + \alpha_1 \log(X_{i,1}) + \alpha_2 X_{i,2}$.
- Zero model: $\text{logit}(p_i) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2}$.
- The zero indicator: $O_i \sim \text{Binom}(1, p_i)$.
- The count response: $y_i = (1 - O_i) \times C_i$,
where $C_i \sim \text{Truncated NB}(\mu_i, \gamma, \pi), \gamma = 4$ is the dispersion parameter,
and $\pi = \gamma / (\gamma + \mu_i)$ is the probability.
- The steps are repeated for all simulated datasets. We generate 600 response variables from the HNB model with sample sizes of $n = 50, 100, 200$ and zero proportions of $p = 30\%$.

- **Logarithmic HNB Model (true model):**

$$y_i \sim \text{HNB}(\mu_i, \gamma, p_i),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 \log(X_{i,1}) + \alpha_2 X_{i,2} + \sum_{j=1}^5 a_j Z_{i,j},$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \sum_{j=1}^5 b_j Z_{i,j};$$

- **Linear HNB Model (wrong model):**

$$y_i \sim \text{HNB}(\mu_i, \gamma, p_i),$$

$$\log(\mu_i) = \alpha_0 + \alpha_1 X_{i,1} + \alpha_2 X_{i,2} + \sum_{j=1}^5 a_j Z_{i,j},$$

$$\text{logit}(p_i) = \beta_0 + \beta_1 X_{i,1} + \beta_2 X_{i,2} + \sum_{j=1}^5 b_j Z_{i,j}.$$

- We calculate four different kinds of Z-residuals and compare them.
- Other than SW test, we introduce Analysis of Variance (AOV) test to detect potential non-linear covariate effect by comparing the group means of Z-residuals.
- This simulation study focuses on comparing the performance between the SW test and AOV test based on RPP Z-residuals.
- We provide both graphical (scatter plots, box plots) and numerical (SW test for checking normality, AOV test for checking homogeneity) results for Z-residuals.

Subsection 2

Z-residual Diagnostics

Posterior RPP Z-residual: Scatter Plot

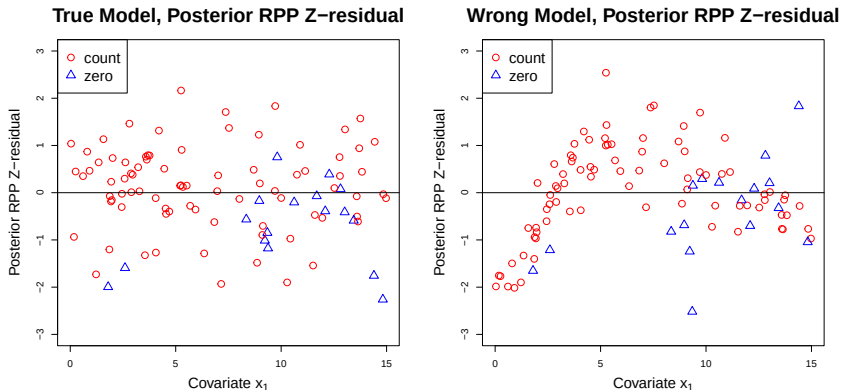


Figure 5: Posterior RPP Z-residuals vs. covariate $X_{i,1}$ for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100, p = 30\%$).

Posterior RPP Z-residual: Q-Q Plot

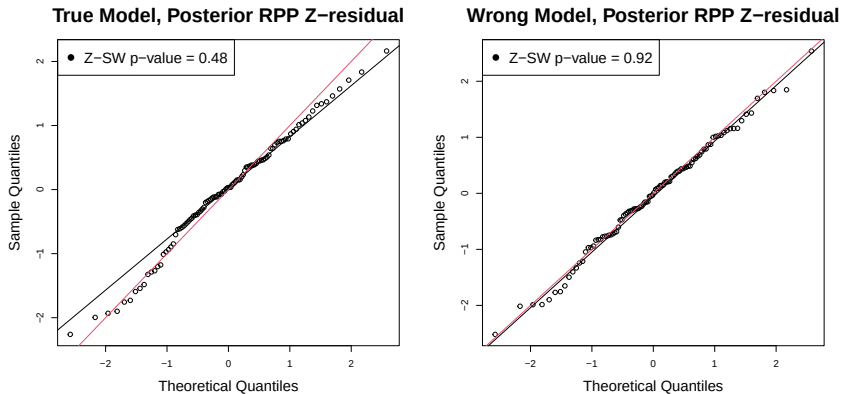


Figure 6: Q-Q plots for posterior RPP Z-residuals for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100, p = 30\%$).

SW Test vs. AOV Test: Box Plot

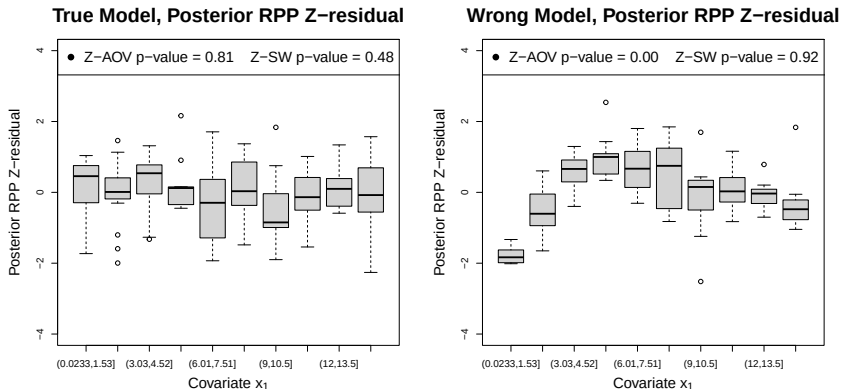


Figure 7: Box plots of posterior RPP Z-residuals for logarithmic HNB and linear HNB models when the dataset is simulated from logarithmic HNB ($n = 100, p = 30\%$). The p -values for the SW test and AOV test are shown on the top.

Subsection 3

Type-I Error Rates and Powers of SW and AOV Tests

Model Rejection Rates

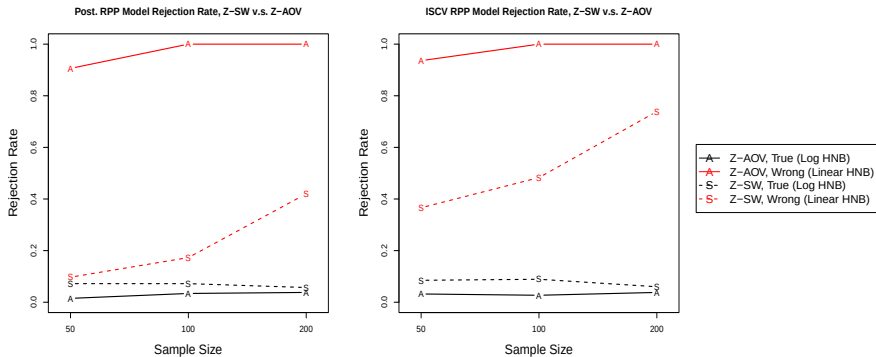


Figure 8: Model rejection rates of RPP Z-residual-based SW (Z-SW) Tests versus RPP Z-residual-based AOV (Z-AOV) Tests when the dataset is simulated from logarithmic HNB. A model is rejected when the test p -value is smaller than 5%. The number of replications is 600. The zero proportion is $p = 30\%$.

Section 6

Conclusions and Future Work

Conclusions and Future Work

- We have developed generic R functions to compute these four types of Z-residuals for Bayesian hurdle models.
- The performance of RPP Z-residuals is better than MPP Z-residuals in identifying wrong distributional families, as indicated by the type I error rates and statistical power.
- The performance of Z-AOV test is better than the Z-SW test in detecting the functional form of covariates, based on the type I error rates and statistical power.
- For future work, we are interested in comparing the difference between posterior and ISCV Z-residuals in discerning true and wrong models.

References

- Li, L., Wu, T., & Feng, C. (2021). Model diagnostics for censored regression via randomized survival probabilities. *Statistics in medicine*, 40(6), 1482-1497.
- Li, L., Qiu, S., Zhang, B., & Feng, C. X. (2016). Approximating cross-validators predictive evaluation in Bayesian latent variable models with integrated IS and WAIC. *Statistics and Computing*, 26, 881-897.
- Li, L., Feng, C. X., & Qiu, S. (2017). Estimating cross-validators predictive p-values with integrated importance sampling for disease mapping models. *Statistics in medicine*, 36(14), 2220-2236.
- Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and computing*, 27, 1413-1432.
- Feng, C., Li, L., & Sadeghpour, A. (2020). A comparison of residual diagnosis tools for diagnosing regression models for count data. *BMC Medical Research Methodology*, 20(1), 1-21.

Thank you!
Welcome questions :)